

COAL Assignment#1

Total 60 Marks

Section: BDS (A, B, C)

Deadline: 21 September 2022

Submit hardcopy in class and soft copy in google classroom.

Q1.

Identify the problems in the following instructions and correct them by replacing them with one or two instructions having the same effect. [05 marks]

a. mov [02], [22]

b. mov [wordvar], 20

c. mov bx, al

d. mov ax, [si+di+100]

Q2.

Write a program that determines the maximum magnitude value in a signed array of given size. Use the data definitions given as under, where label 'array' is the array itself, 'size' is the count of elements in the array. The program should put the maximum value found at label 'max'. For example, the maximum magnitude value in the following data of 7 elements is -10 (not 6) because magnitude of -10 i.e. 10 is greater than the magnitude of 6. [10 Marks]

array: db -3,5,-10,4,6,-7,1

size: db 7

max: db 0

Q3.

Calculate the physical memory address generated by the following segment offset pairs. [07 Marks]

a. 1DDD:0436

b. 1234:7920

c. 74F0:2123

d. 0000:6727

e. FFFF:4336

f. 1080:0100

g. AB01:FFFF

Q4.

What are the first and the last physical memory addresses accessible using the following segment values? [05 Marks]

- a. 1000
- b. 0FFF
- c. 1002
- d. 0001
- e. E000

Q5.

What is the effective address generated by the following instructions? Every instruction is independent of others. Initially BX=0x0100, num1=0x1001, [num1]=0x0000, and SI=0x0100 [08 Marks]

- a. mov ax, [bx+12]
- b. mov ax, [bx+num1]
- c. mov ax, [num1+bx]
- d. mov ax, [bx+si]

Q6.

What is the effective address generated by the following combinations if they are valid. If not give reason. Initially BX=0x0100, SI=0x0010, DI=0x0001, BP=0x0200, and SP=0xFFFF [06 Marks]

- a. bx-si
- b. bx-bp
- c. bx+10
- d. bx-10
- e. bx+sp
- f. bx+di

Q7.

Consider the program written below. For this program being executed the current contents of CS register are 0FE2h and the Physical address pointed to by CS:IP is 0FF29. Size of mov instruction is 3 bytes. For this program, determine the current contents of a) IP register, and b) ax register. [04 Marks]

```
[ORG 0X100]
MOV AX, 10
MOV AX, 6
MOV AX, 5
MOV AX, 2
MOV AX, 1
MOV AX, 0x4C00
INT 0x21
```

Q8.

Machine Code from a listing file is given below. A1 is the opcode to move contents from memory to register ax. Determine the Physical address referred to by each instruction. The contents of segment registers are: CS= 00E2h, DS= 0FE2h, SS= 0FF0h, ES= 0FFFh. [05Marks]

- i. A1[1700]
- ii. A1[120B]

Q9.

Write a program to multiply two 32bit numbers and store the answer in a 64bit location [10 Marks]

Q10. Given the code below, answer the following questions [5x3 Marks]:

[org 0x100]	19. cmp si, di
1. mov si, -1	20. jnl end
2. mov di, [size]	21. mov al, [arr+si]
3.	22. mov ah, [arr+di]
4. l1:	23. mov [arr+di], al
5. add si, 1	24. mov [arr+si], ah
6. cmp si, di	25. jmp l1
7. je end	26. end:
8. mov al, [arr+si]	27.
9. shr al, 1	28. mov ax, 0x4c00
10. jnc l1	29. int 21h
11.	30. arr: db 3, 10, 2, 0, 7, 5, 8
12. l2:	31. size: dw 7
13. sub di, 1	
14. cmp si, di	
15. je end	
16. mov al, [arr+di]	
17. shr al, 1	
18. jc l2	

1) If the label 'size' is at 0x0149 what is the total size of program in bytes (decimal value) excluding data (i.e. arr label)?

2) What changes will be made to 'arr' after the program execution? Fill the table below:

Before program:	arr:	3	10	2	0	7	5	8
After program:	arr:							

3) Rewrite the above code from line 4 to 10 without using shr.